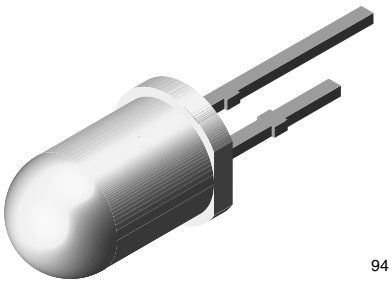
Ambient Light Sensor



94 8390

FEATURES

- Package type: leaded
- Package form: T-1¾
- Dimensions (in mm): Ø 5
- High photo sensitivity
- Adapted to human eye responsivity
- Angle of half sensitivity: φ = ± 20°
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

Pb-free

e4

RoHS COMPLIANT

HALOGEN FREE GREEN (5-2008)

DESCRIPTION

TEPT5600 ambient light sensor is a silicon NPN epitaxial planar phototransistor in a T-1¾ package. It is sensitive to visible light much like the human eye and has peak sensitivity at 570 nm.

APPLICATIONS

- Replacement of cadmium sulfide (CdS) photoresistors
- Ambient light sensor

PRODUCT SUMMARY			
COMPONENT	I_{PCE} (μA)	φ (deg)	$\lambda_{0.5}$ (nm)
TEPT5600	630	± 20	440 to 800

Note

- Test condition see table “Basic Characteristics”

ORDERING INFORMATION			
ORDERING CODE	PACKAGING	REMARKS	PACKAGE FORM
TEPT5600	Bulk	MOQ: 4000 pcs, 4000 pcs/bulk. Label with I_{PCE} group on each bulk. Specifications of group A/B/C/D see table “Type Dedicated Characteristics”	T-1¾

Note

- MOQ: minimum order quantity

ABSOLUTE MAXIMUM RATINGS (T_{amb} = 25 °C, unless otherwise specified)				
PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
Collector emitter voltage		V_{CEO}	6	V
Emitter collector voltage		V_{ECO}	1.5	V
Collector current		I_C	20	mA
Power dissipation	$T_{amb} \leq 55$ °C	P_V	100	mW
Junction temperature		T_j	100	°C
Operating temperature range		T_{amb}	-40 to +85	°C
Storage temperature range		T_{stg}	-40 to +100	°C
Soldering temperature	t ≤ 3 s, 2 mm distance to package	T_{sd}	260	°C
Thermal resistance junction/ambient	J-STD-051, soldered on PCB	R_{thJA}	230	K/W

For technical questions, contact: detectortechsupport@vishay.com

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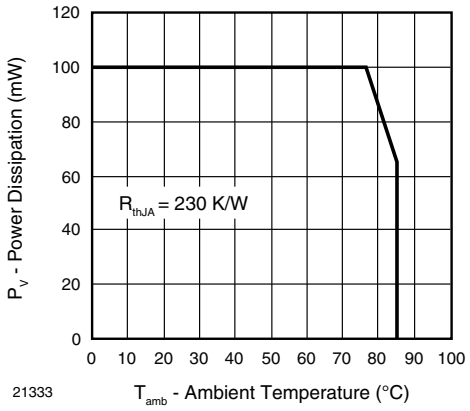


Fig. 1 - Power Dissipation Limit vs. Ambient Temperature

BASIC CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)						
PARAMETER	**TEST CONDITION**	**SYMBOL**	**MIN.**	**TYP.**	**MAX.**	**UNIT**
Collector emitter breakdown voltage	I_C = 0.1 mA	V_{CEO}	6			V
Collector dark current	V_{CE} = 5 V, E = 0	I_{CEO}		3	50	nA
Collector emitter capacitance	V_{CE} = 0 V, f = 1 MHz, E = 0	C_{CEO}		16		pF
Photo current	E_v = 20 lx, CIE illuminant A, V_{CE} = 5 V	I_{PCE}	25		226.8	µA
	E_v = 100 lx, CIE illuminant A, V_{CE} = 5 V	I_{PCE}		630		µA
Angle of half sensitivity		φ		± 20		deg
Wavelength of peak sensitivity		λ_p		570		nm
Range of spectral bandwidth		$\lambda_{0.5}$		440 to 800		nm

TYPE DEDICATED CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)						
PARAMETER	**TEST CONDITION**	**BINNED GROUP**	**SYMBOL**	**MIN.**	**MAX.**	**UNIT**
Photo current	E_V = 20 lx, CIE illuminant A, V_{CE} = 5 V, T_{amb} = 25 °C	A	I_{PCE}	25	50.4	µA
		B	I_{PCE}	41.7	84	µA
		C	I_{PCE}	69.4	140	µA
		D	I_{PCE}	113.4	226.8	µA

Note

- Each 4000 piece bag will contain a single group. The label on the bag will indicate which binned group is in the bag. A specific group cannot be ordered. Production shipments containing multiple bags will likely include multiple groups. Please design accordingly.

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BASIC CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)

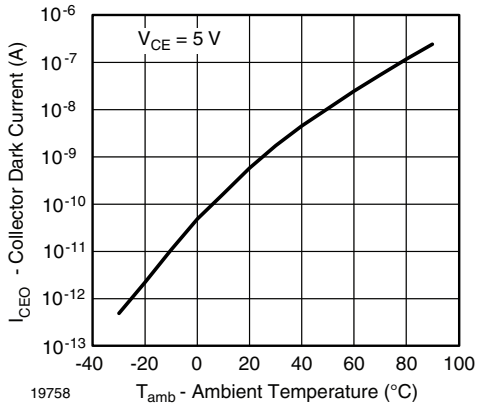


Fig. 2 - Collector Dark Current vs. Ambient Temperature

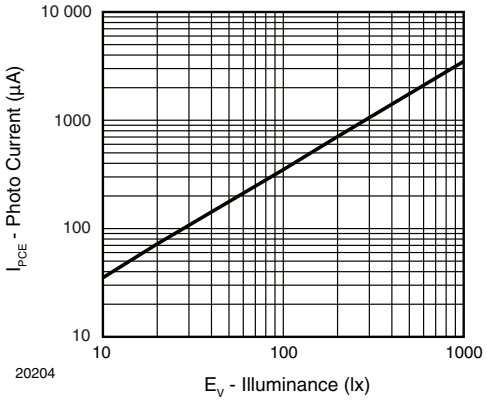


Fig. 5 - Photo Current vs. Illuminance

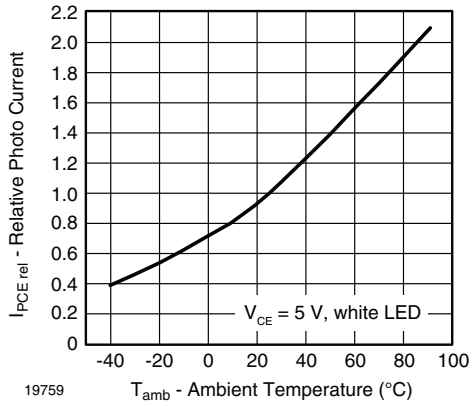


Fig. 3 - Relative Photo Current vs. Ambient Temperature

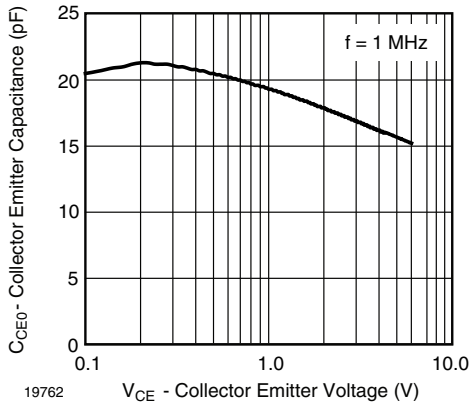


Fig. 6 - Collector Emitter Capacitance vs. Collector Emitter Voltage

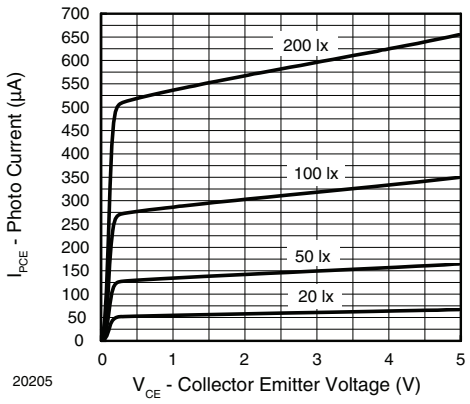


Fig. 4 - Photo Current vs. Collector Emitter Voltage

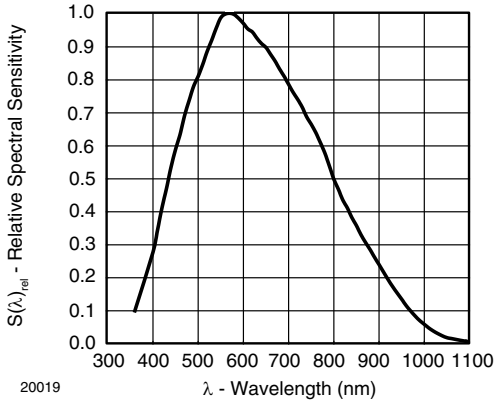


Fig. 7 - Relative Spectral Sensitivity vs. Wavelength

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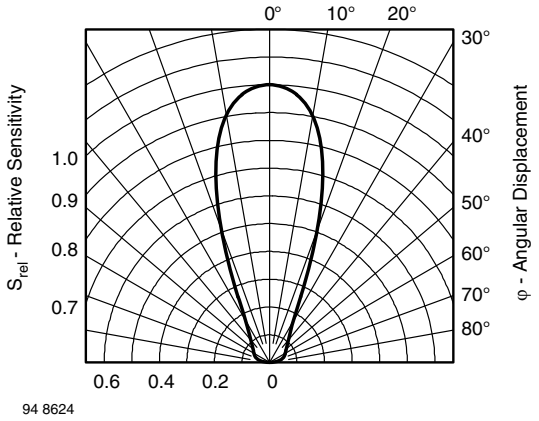
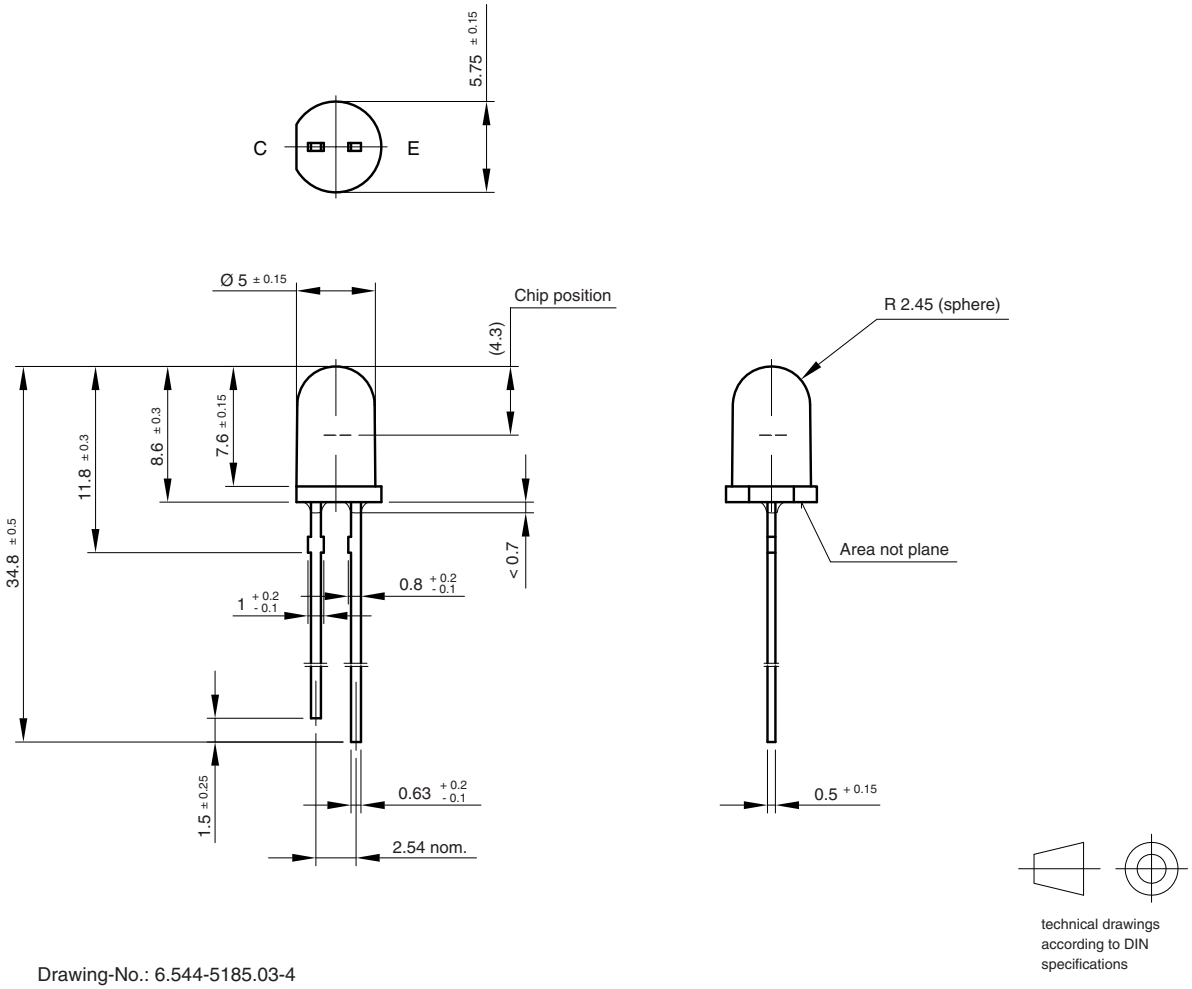


Fig. 8 - Relative Radiant Sensitivity vs. Angular Displacement

PACKAGE DIMENSIONS in millimeters



Drawing-No.: 6.544-5185.03-4
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